HW6 阮柏瀚 數值方法 E84116049

1.
$$\begin{bmatrix} 1.19 & 2.11 & -100 & 1 \\ 14.2 & -0.112 & 12.2 & -1 \\ 0 & 100 & -99.9 & 1 \\ 15.3 & 0.11 & -13.1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.12 \\ 3.44 \\ 2.15 \\ 4.16 \end{bmatrix}$$

$$\begin{bmatrix} 15.3 & 0.1 & -13.1 & -1 \\ 0 & 100 & -99.9 & 1 \\ 0 & 0 & 24.1443 & -0.06976 \\ 0 & 0 & 0 & 0.7769 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.16 \\ 2.15 \\ -0.4163 \\ -0.9193 \end{bmatrix}$$

$$x_4 = \frac{-0.9193}{0.7769} = -1.1833 \quad \#$$

$$\begin{bmatrix} 15.3 & 0.11 & -13.1 & -1 \\ 14.2 & -0.112 & 12.2 & -1 \\ 0 & 100 & -99.9 & 1 \\ 1.19 & 2.11 & -100 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.16 \\ 3.44 \\ 2.15 \\ 1.12 \end{bmatrix}$$

$$24.1443 x_3 - 0.06976 x_4 = -0.4163$$

$$x_3 = -0.02066 \quad \#$$

$$\begin{bmatrix} 15.3 & 0.11 & -13.1 & -1 \\ 0 & -0.2141 & 24.3582 & -0.0719 \\ 0 & 100 & -99.9 & 1 \\ 0 & 2.1014 & -98.9811 & 1.0778 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.16 \\ -0.4209 \\ 2.15 \\ 0.7964 \end{bmatrix}$$

$$100 x_2 - 99.9 x_3 + 1 x_4 = 2.15$$

$$x_2 = 0.01269 \quad \#$$

$$15.3 x_1 + 0.1 x_2 - 13.1 x_3 - 1 x_4 = 4.16$$

$$x_1 = 0.1768 \quad \#$$

$$\begin{bmatrix} 15.3 & 0.11 & -13.1 & -1 \\ 0 & 100 & -99.9 & 1 \\ 0 & -0.2141 & 24.3582 & -0.0719 \\ 0 & 2.1014 & -98.9811 & 1.0778 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.16 \\ 2.15 \\ -0.4209 \\ 0.7964 \end{bmatrix}$$

$$\begin{bmatrix} 15.3 & 0.1 & -13.1 & -1 \\ 0 & 100 & -99.9 & 1 \\ 0 & 0 & 24.1443 & -0.06976 \\ 0 & 0 & -96.8818 & 1.0568 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4.16 \\ 2.15 \\ -0.4163 \\ 0.7512 \end{bmatrix}$$

2.

$$A=\begin{bmatrix} 4 & 1 & -1 & 0 \\ 1 & 3 & -1 & 0 \\ -1 & -1 & 6 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix} = L\ U = \begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ 0 & l_{42} & l_{43} & l_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} & 0 \\ 0 & 1 & u_{23} & u_{24} \\ 0 & 0 & 1 & u_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

① $l_{11}=4$

$l_{21}=1$

$l_{31}=-1$

② $l_{11}u_{12}=1$

$u_{12}=0.25$

$l_{11}u_{13}=-1$

$u_{13}=-0.25$

③ $l_{21}u_{12}+l_{22}=3$

$l_{22}=\frac{11}{4}$

$l_{31}u_{12}+l_{32}=-1$

$l_{32}=-\frac{3}{4}$

$l_{42}=0$

④

$l_{21}u_{13}+l_{22}u_{23}=-1$

$u_{23}=-\frac{3}{11}$

$l_{22}u_{24}=0$

$u_{24}=0$

⑤

$l_{31}u_{13}+l_{32}u_{23}+l_{33}\times 1=6$

$l_{33}=\frac{61}{11}$

$l_{42}u_{23}+l_{43}=2$

$l_{43}=2$

⑥

$l_{32}u_{24}+l_{33}u_{34}=2$

$u_{34}=\frac{22}{61}$

⑦ $l_{42}u_{24}+u_{34}l_{43}+l_{44}=5$

$l_{44}=\frac{261}{61}$

$$\begin{array}{cc} L & U \\ \begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & \frac{11}{4} & 0 & 0 \\ -1 & -\frac{3}{4} & \frac{61}{11} & 0 \\ 0 & 0 & 2 & \frac{261}{61} \end{bmatrix} & \begin{bmatrix} 1 & \frac{1}{4} & \frac{-1}{4} & 0 \\ 0 & 1 & -\frac{3}{11} & 0 \\ 0 & 0 & 1 & \frac{22}{61} \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

$L\vec{y}=b$

$A\cdot A^{-1}=1$ $\quad U\vec{x}=\vec{y}$

$$\begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & \frac{11}{4} & 0 & 0 \\ -1 & -\frac{3}{4} & \frac{61}{11} & 0 \\ 0 & 0 & 2 & \frac{261}{61} \end{bmatrix} \begin{bmatrix} y_{11} \\ y_{21} \\ y_{31} \\ y_{41} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y_{11} = 0.25$$

$$y_{11} + \frac{11}{4} y_{21} = 0$$

$$y_{21} = \frac{-1}{11}$$

$$-y_{11} - \frac{3}{4} y_{21} + \frac{61}{11} y_{31} = 0$$

$$y_{31} = \frac{2}{61}$$

$$2y_{31} + \frac{261}{61} y_{41} = 0$$

$$y_{41} = -\frac{4}{261}$$

$$\begin{bmatrix} 1 & \frac{1}{4} & \frac{-1}{4} & 0 \\ 0 & 1 & -\frac{3}{11} & 0 \\ 0 & 0 & 1 & \frac{22}{61} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \\ x_{31} \\ x_{41} \end{bmatrix} = \begin{bmatrix} 0.25 \\ \frac{-1}{11} \\ \frac{2}{61} \\ -\frac{4}{261} \end{bmatrix}$$

$$x_{41} = -0.0153$$

$$x_{31} + \frac{22}{61} \times -\frac{4}{261} = \frac{2}{61}$$

$$x_{31} = 0.0383$$

$$x_{21} + -\frac{3}{11} x_{31} = \frac{-1}{11}$$

$$x_{21} = -0.0805$$

$$x_{11} + \frac{1}{4} x_{21} + \frac{-1}{4} x_{31} = \frac{1}{4}$$

$$x_{11} = 0.2797$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$y_{12} = 0$$

$$y_{12} + \frac{11}{4} y_{22} = 1$$

$$y_{22} = \frac{4}{11}$$

$$-y_{12} - \frac{3}{4} y_{22} + \frac{61}{11} y_{32} = 0$$

$$y_{32} = 0.0492$$

$$2y_{32} + \frac{261}{61} y_{42} = 0$$

$$y_{42} = -0.023$$

$$\begin{bmatrix} 0 \\ \frac{4}{11} \\ 0.0492 \\ -0.023 \end{bmatrix}$$

$$x_{42} = -0.023$$

$$x_{32} + \frac{22}{61} \times x_{42} = 0.0492$$

$$x_{32} = 0.0575$$

$$x_{22} + -\frac{3}{11} x_{32} = \frac{4}{11}$$

$$x_{22} = 0.3792$$

$$x_{12} + \frac{1}{4} x_{22} + \frac{-1}{4} x_{32} = 0$$

$$x_{12} = -0.0804$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$y_{13} = 0$$

$$y_{23} = 0$$

$$-1 y_{13} - \frac{3}{4} y_{23} + \frac{61}{11} y_{33} = 1$$

$$y_{33} = \frac{11}{61}$$

$$2y_{33} + \frac{261}{61} y_{43} = 0$$

$$y_{43} = -\frac{22}{261}$$

$$\begin{bmatrix} 0 \\ 0 \\ \frac{11}{61} \\ \frac{-22}{261} \end{bmatrix}$$

$$x_{43} = -0.0843$$

$$x_{33} + \frac{22}{61} x_{43} = \frac{11}{61}$$

$$x_{33} = 0.2107$$

$$x_{23} - \frac{3}{11} x_{33} = 0$$

$$x_{23} = 0.0575$$

$$x_{13} + \frac{1}{4} x_{23} + \frac{-1}{4} x_{33} = 0$$

$$x_{13} = 0.0383$$

$$
\begin{array}{c}
\\
\begin{bmatrix} 1 & \frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 1 & -\frac{3}{11} & 0 \\ 0 & 0 & 1 & \frac{22}{61} \\ 0 & 0 & 0 & 1 \end{bmatrix}
\begin{bmatrix} y_{11} & \cdots & y_{14} \\ y_{21} & & \vdots \\ y_{31} & & \vdots \\ y_{41} & \cdots & y_{44} \end{bmatrix}
=
\end{array}
\begin{array}{c}
\begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\
\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{array}
$$

①

$y_{41} = 0$

$y_{31} + \frac{22}{61} y_{41} = 0$

$y_{31} = 0$

$y_{21} + \frac{-3}{11} y_{31} = 0$

$y_{21} = 0$

$y_{11} = 1$

$$
\begin{bmatrix} 4 & 0 & 0 & 0 \\ 1 & \frac{11}{4} & 0 & 0 \\ -1 & -\frac{3}{4} & \frac{61}{11} & 0 \\ 0 & 0 & 2 & \frac{261}{61} \end{bmatrix}
\begin{bmatrix} A^{-1}_{11} \\ A^{-1}_{21} \\ A^{-1}_{31} \\ A^{-1}_{41} \end{bmatrix}
=
\begin{bmatrix} \\ \\ \\ \end{bmatrix}
$$

$4A_{11}$

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$y_{14} = 0$

$y_{24} = 0$

$y_{34} = 0$

$y_{44} = \frac{61}{261}$

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{61}{261} \end{bmatrix}$

$x_{44} = 0.2337$

$x_{34} + \frac{22}{61} x_{44} = 0$

$x_{34} = -0.0843$

$x_{24} - \frac{3}{11} x_{34} = 0$

$x_{24} = -0.023$

$x_{14} + \frac{1}{4} x_{24} + \frac{-1}{4} x_{34} = 0$

$x_{14} = -0.0153$

$$
A^{-1} = \begin{bmatrix} 0.2797 & -0.0804 & 0.0383 & -0.0153 \\ -0.0805 & 0.3792 & 0.0575 & -0.023 \\ 0.0383 & 0.0575 & 0.2107 & -0.0843 \\ -0.0153 & -0.023 & -0.0843 & 0.2337 \end{bmatrix}
$$

#

3.

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} = \begin{bmatrix} \overset{①}{l_{11}} & 0 & 0 & 0 \\ l_{21} & \overset{③}{l_{22}} & 0 & 0 \\ 0 & l_{32} & \overset{⑤}{l_{33}} & 0 \\ 0 & 0 & l_{43} & \overset{⑦}{l_{44}} \end{bmatrix} \begin{bmatrix} 1 & \overset{②}{u_{12}} & 0 & 0 \\ 0 & 1 & \overset{④}{u_{23}} & 0 \\ 0 & 0 & 1 & \overset{⑥}{u_{34}} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

① $l_{11} = 3$

$l_{21} = -1$

② $l_{11} \cdot u_{12} + 0 \times 1 = -1$

$u_{12} = \frac{-1}{3}$

③ $l_{21} \cdot u_{12} + l_{22} \times 1 = 3$

$l_{22} = \frac{8}{3}$

$l_{32} = -1$

④ $l_{22} u_{23} + 0 \times 1 = -1$

$u_{23} = \frac{-3}{8}$

⑤ $l_{32} u_{23} + l_{33} 1 = 3$

$l_{33} = \frac{21}{8}$

$l_{43} = -1$

⑥ $l_{33} u_{34} + 0 \times 1 = -1$

$u_{34} = \frac{-8}{21}$

⑦ $l_{43} u_{34} + l_{44} = 3$

$l_{44} = \frac{55}{21}$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ -1 & \frac{8}{3} & 0 & 0 \\ 0 & -1 & \frac{21}{6} & 0 \\ 0 & 0 & -1 & \frac{55}{21} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

$y_1 = \frac{2}{3}$

$-1 y_1 + \frac{8}{3} y_2 = 3$

$y_2 = 1.375$

$-1 y_2 + \frac{21}{8} y_3 = 4$

$y_3 = \frac{43}{21}$

$-1 y_3 + \frac{55}{21} y_4 = 1$

$y_4 = \frac{64}{55}$

$$\begin{bmatrix} 1 & -\frac{1}{3} & 0 & 0 \\ 0 & 1 & \frac{-3}{8} & 0 \\ 0 & 0 & 1 & \frac{-8}{21} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ 1.375 \\ \frac{43}{21} \\ \frac{64}{55} \end{bmatrix}$$

$x_4 = \frac{64}{55}$

$x_3 + \frac{-8}{21} x_4 = \frac{43}{21}$

$x_3 = \frac{137}{55}$

$x_2 + \frac{-3}{8} x_3 = 1.375$

$x_2 = \frac{127}{55}$

$x_1 + \frac{-1}{3} x_2 = \frac{2}{3}$

$x_1 = \frac{79}{55}$

L U

$$A = \begin{bmatrix} 3 & 0 & 0 & 0 \\ -1 & \frac{8}{3} & 0 & 0 \\ 0 & -1 & \frac{21}{8} & 0 \\ 0 & 0 & -1 & \frac{55}{21} \end{bmatrix} \begin{bmatrix} 1 & \frac{-1}{3} & 0 & 0 \\ 0 & 1 & \frac{-3}{8} & 0 \\ 0 & 0 & 1 & \frac{-8}{21} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$L \vec{y} = b$

$U \vec{x} = \vec{y}$

$x_1 = \frac{79}{55}$

$x_2 = \frac{127}{55}$

$x_3 = \frac{137}{55}$

$x_4 = \frac{64}{55}$ #